

Gauge-Complete Reconstruction of the Physical Quartic Divisor

Center, RMS Radius, Shape, and the Next Gate-A Route

Liang Wang

July 2026

Abstract

Centered-RMS normalization exposes the exact two-coordinate quartet shape, but it discards spectral center and scale. This paper restores those data and proves a complete affine-gauge dictionary for every nondegenerate conjugate quartet.

Let μ be the root barycenter, $r > 0$ the centered RMS radius, and $Q_{u,\eta}$ the canonical shape polynomial of RH-213. The raw monic divisor is exactly

$$D(z) = r^4 Q_{u,\eta}\left(\frac{z - \mu}{r}\right) = (z - \mu)^4 + c_2 r^2 (z - \mu)^2 + c_3 r^3 (z - \mu) + c_4 r^4.$$

Thus (μ, r, u, η) reconstructs both the root multiset and every raw coefficient. Conversely these parameters are recovered from the roots, up to the already classified branch-label fibers on the shape boundary. Shape alone is non-injective: every translation and positive dilation orbit has the same (u, η) .

All 32 physical endpoint quartets are reconstructed with maximum root error 3.15×10^{-16} and coefficient error 7.78×10^{-16} . An exact gauge-first, shape-second telescoping decomposition is audited on all 30 adjacent channel transitions; eight are gauge-leg dominant and 22 are shape-leg dominant. Neither ledger can be discarded.

Together with RH-219's fixed-degree obstruction, this yields a route decision: the next Gate-A object must be a rank-growing, gauge-complete physical divisor using one global center and radius per cloud. The present paper closes finite reconstruction only; no growing determinant or Gate-A closure is obtained.

1 Why the quotient must be completed

RH-212 showed that centered-RMS coefficients do not contract more strongly than the raw quartic, while RH-213 proved that they lie on an exact two-dimensional shape manifold [1, 2]. The quotient is valuable because it separates geometry, but it cannot by itself reconstruct the raw spectral factor.

RH-219 further proves that one fixed quartic, even repeated, cannot furnish a locally finite growing spectrum [3]. Before moving to larger clouds, the finite factor needs a complete state type: not only shape, but also the affine gauge removed by normalization.

2 Gauge and shape data

Let $\Lambda = \{\lambda_1, \dots, \lambda_4\}$ be a nonconstant conjugate-closed quartet. Define

$$\mu = \frac{1}{4} \sum_{j=1}^4 \lambda_j, \quad r = \left(\frac{1}{4} \sum_{j=1}^4 |\lambda_j - \mu|^2 \right)^{1/2} > 0. \quad (1)$$

Conjugate closure makes $\mu \in \mathbb{R}$. The normalized roots

$$q_j = \frac{\lambda_j - \mu}{r} \quad (2)$$

are centered and have unit mean square modulus. With the branch convention of RH-213 they determine (u, η) and

$$Q_{u,\eta}(w) = w^4 + c_2 w^2 + c_3 w + c_4, \quad (3)$$

where

$$c_2 = 2 - 4u, \text{quad} c_3 = 4\sqrt{u}(1 - u)\eta, \text{quad} c_4 = 1 - \eta^2(1 - u)^2. \quad (4)$$

3 Exact reconstruction theorem

Theorem 3.1 (Gauge-complete divisor formula). *The monic raw divisor of Λ is*

$$D(z) = \prod_{j=1}^4 (z - \lambda_j) = r^4 Q_{u,\eta} \left(\frac{z - \mu}{r} \right). \quad (5)$$

Equivalently,

$$D(z) = (z - \mu)^4 + c_2 r^2 (z - \mu)^2 + c_3 r^3 (z - \mu) + c_4 r^4. \quad (6)$$

Proof. Since $\lambda_j = \mu + r q_j$,

$$\prod_j (z - \lambda_j) = \prod_j \left(r((z - \mu)/r - q_j) \right) = r^4 Q_{u,\eta}((z - \mu)/r). \quad (7)$$

Substituting (3) gives (6). $\square$

Expanding in powers of z gives the explicit raw coefficient dictionary:

$$\begin{aligned} D(z) = & z^4 - 4\mu z^3 + (6\mu^2 + c_2 r^2) z^2 \\ & + (-4\mu^3 - 2\mu c_2 r^2 + c_3 r^3) z \\ & + (\mu^4 + \mu^2 c_2 r^2 - \mu c_3 r^3 + c_4 r^4). \end{aligned} \quad (8)$$

Corollary 3.2 (Complete finite state type). *Away from the branch-label boundary fibers classified in RH-213, the four real parameters (μ, r, u, η) determine and are determined by a branch-labeled conjugate quartet.*

Proof. The forward direction is Theorem 3.1. Conversely the roots give μ and r by (1); normalized branch geometry gives u and η . $\square$

The parameter count matches the four real coefficients of a generic monic real quartic.

4 Shape alone is non-injective

Proposition 4.1 (Affine gauge fibers). *Fix an interior shape (u, η) . The family*

$$\Lambda_{\mu, r} = \{\mu + rq_j(u, \eta) : 1 \leq j \leq 4\}, \quad \mu \in \mathbb{R}, \quad r > 0, \quad (9)$$

has the same centered shape for every (μ, r) but consists of distinct raw divisors unless the gauge pairs coincide.

Proof. Centering and RMS normalization remove μ and r , so the shape is fixed. If μ changes, the cubic coefficient -4μ changes. With μ fixed, a change in r changes root distances from the center and hence the divisor. $\square$

Therefore a normalized shape limit cannot be silently identified with a raw determinant limit. A growing construction must retain a gauge ledger.

5 Exact transition decomposition

Let

$$\theta_0 = (\mu_0, r_0, u_0, \eta_0), \quad \theta_1 = (\mu_1, r_1, u_1, \eta_1). \quad (10)$$

Insert the mixed state $\theta_\times = (\mu_1, r_1, u_0, \eta_0)$. Then

$$C(\theta_1) - C(\theta_0) = \underbrace{C(\theta_\times) - C(\theta_0)}_{\text{gauge leg}} + \underbrace{C(\theta_1) - C(\theta_\times)}_{\text{shape leg}}, \quad (11)$$

where C is the raw coefficient vector from (8).

This path is ordered—gauge first, shape second—so its two norms are not an orthogonal variance decomposition. It is nevertheless exact and answers a useful diagnostic question: can raw motion be explained by gauge alone or shape alone on each transition?

6 Physical reconstruction audit

The sixteen-level, two-channel atlas supplies 32 quartets and 30 adjacent channel transitions. Direct root expansion is compared with Theorem 3.1.

diagnostic	maximum
root multiset reconstruction error	3.15×10^{-16}
raw coefficient reconstruction error	7.78×10^{-16}
parameter recovery error	floating zero
path telescoping residual	1.39×10^{-17}

Of the 30 ordered transition decompositions, eight have a larger gauge leg and 22 have a larger shape leg. Thus the failure of centered normalization to contract in RH-212 is not surprising: on most transitions genuine shape motion is at least as large as the discarded gauge motion. The eight gauge-dominant cases also show why raw coefficients cannot be replaced by shape coordinates alone.

7 Dual-channel gauge coherence

At the coarsest level the channel center discrepancy is about 0.00204 and the RMS-radius discrepancy 0.00077. At the finest level they fall to approximately 1.06×10^{-5} and 3.60×10^{-4} . Shape discrepancies remain comparably small. These finite observations support a shared gauge-complete cloud experiment, but do not prove uniform channel convergence.

8 Route decision after the fixed-factor obstruction

The combined state of Gate A is now:

1. the local quartet has canonical branch labels and dual-channel coherence;
2. its affine quotient and boundary geometry are exact;
3. no predictive fixed-shape semigroup has been identified;
4. a fixed or repeated quartic cannot provide a locally finite growing spectrum;
5. the raw factor is exactly recoverable when gauge data are retained.

The next object should therefore be a rank-growing cloud

$$\Lambda_{\sigma,k} = \{\lambda_{\sigma,1}, \dots, \lambda_{\sigma,k}\}, \quad k \rightarrow \infty, \quad (12)$$

with one global center $\mu_{\sigma,k}$ and one global RMS radius $r_{\sigma,k}$ for the entire cloud. Independently normalizing every quartet would destroy relative location between factors and cannot define one canonical product.

The route coordinate is

$$\boxed{\text{finite_gauge_complete_shape_flow_open_rank_growing_divisor}}.$$

9 Claim boundary

This paper closes finite algebraic reconstruction. It does not prove that the gauge parameters converge, construct the rank-growing physical cloud, control a canonical product, or realize a Fredholm determinant. Gate A is open. Gates B–E, Hilbert–Pólya, zeta-zero identification, and RH remain untouched.

References

- [1] Liang Wang. An intrinsic normalization audit for the physical quartic divisor, 2026. RH-212 preprint.
- [2] Liang Wang. The centered conjugate-quartet shape manifold, 2026. RH-213 preprint.
- [3] Liang Wang. Why a fixed quartic cannot supply a growing spectral count, 2026. RH-219 preprint.